CE 310 — Week 7 Assignment: Correlation — Extended Analysis

Due: Wednesday of Week 8 by 9:00 AM — submit to D2L | Total: 60 pts | Dataset: Week5_DOE_MedOffice_ZoneTemp_2023.csv (same file, 8,760 rows)

Type all written responses in the designated yellow cells of your Assignment Answer Sheet. Submit `CE310_W7_<NetID>.xlsx` to D2L.

How to Submit

> Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

Requirement	Detail
File format	`.xlsx` only — written responses go in the Answer Sheet cells, not in a separate PDF
Sheet name	Must be exactly `Answers` — do not rename or delete it
Column layout	Enter answers in **Column B** only — do not insert, delete, or move columns
File name	`CE310_W7_<NetID>.xlsx` (e.g. `CE310_W7_abc123.xlsx`)
Submission	Upload the `.xlsx` file to D2L Dropbox before the deadline

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Problem 1 — New Correlation Computations (30 pts)

These three questions require computations **not performed in the lab**. The lab computed correlations for full-year all-hours, all four seasons (outdoor-cooling), business/non-business hours, non-zero cooling hours, month vs. outdoor/cooling, and covariances. The questions below require different subsets or variable pairs.

P1a (10 pts): Compute r(Outdoor_Temp_F, Cooling_kWh) for **July only** (Month = 7).

Enter your answer to 4 decimal places in the answer sheet.

The lab used seasons (B1–B4). This question uses a single calendar month, giving a higher-resolution view. July is Tucson's hottest and wettest monsoon month — does the outdoor-cooling correlation strengthen or weaken compared to the full-summer r from B1? Explain in 1–2 sentences below your formula in the answer sheet.

P1b (10 pts): Compute r(Zone_Temp_F, Cooling_kWh) for **business hours only** (BusinessHours = "Yes").

Enter your answer to 4 decimal places.

The lab computed r(Zone, Cooling) for all hours (A3) and r(Zone, Outdoor) for business hours (B10). This question asks for r(Zone, Cooling) during business hours only — a different pair on a different subset. During business hours, is zone temperature a better or worse predictor of cooling load than it is across all hours? Explain in 1–2 sentences.

P1c (10 pts): Compute r^2(Zone_Temp_F, Cooling_kWh) for all 8,760 hours, and provide a written interpretation.

Enter the r^2 value to 4 decimal places.

Written interpretation (2–3 sentences in answer sheet): Compare r^2(Zone_Temp_F, Cooling_kWh) to r^2(Outdoor_Temp_F, Cooling_kWh) from lab A5. What percentage of variance in Cooling_kWh is explained by Zone_Temp_F? What percentage is explained by Outdoor_Temp_F? What does this comparison tell a building operator about which sensor is more informative for predicting cooling load?

Problem 2 — ASHRAE 55 Thermal Comfort and Seasonal Zone Control (18 pts)

P2a (6 pts): Compute r(Zone_Temp_F, Outdoor_Temp_F) for **Summer only** and for **Winter only** separately.

Enter both values to 4 decimal places in the answer sheet.

Note: the lab computed r(Zone, Outdoor) for all hours (A1) and for business hours (B10), but NOT by season. These are new computations.

P2b (6 pts — written): Interpret the summer vs. winter r(Zone, Outdoor) values from P2a.

Write 4–6 sentences addressing all of the following:

1. In summer, r(Zone, Outdoor) is **negative**. This is counterintuitive — why would zone temperature go *down* when outdoor temperature goes *up* in summer? (Hint: the HVAC cooling system runs hardest on the hottest summer days, actively depressing zone temperature. The mechanical cooling counteracts the outdoor temperature signal.)
2. In winter, r(Zone, Outdoor) is strongly positive. When outdoor temperature is cold, the heating system may be less active or in setback mode, allowing zone temperature to move with outdoor temperature.
3. Reference ASHRAE 55-2023: the operative temperature comfort zone for office buildings is approximately 68–79°F. In summer, the HVAC works hardest (negative r) to keep zone temp within this range. In winter, the building is closer to free-running.
4. From Week 6 B10: the 99th percentile zone temperature in summer was 79.5°F — near the upper edge of the ASHRAE 55 comfort zone. Does a negative r(Zone, Outdoor) in summer make you more or less confident that the HVAC will protect against exceedance during extreme heat events?

P2c (6 pts — written): A colleague says: “r 0.75 between zone temperature and outdoor temperature for all hours — that’s pretty strong. So the HVAC doesn’t really control the indoor climate very well.”

Write 3–5 sentences explaining why this reasoning is incomplete. Your answer must address:

1. What r does **not** tell you about the magnitude of temperature exceedances. A correlation of 0.75 confirms that zone and outdoor temperatures move together in direction — but it says nothing about how many degrees the zone temperature changes per degree of outdoor change. The regression slope (, which you will compute in Week 9) captures that magnitude.
2. Cite the Week 5 compression ratio: _outdoor / _zone = 16.92 / 2.92 = 5.8×. The HVAC compresses outdoor swings of ±17°F into zone swings of only ±3°F. A correlation of 0.75 is compatible with strong HVAC control when the slope is very small.
3. Cite lab A1 all-hours result and the Week 5 result that P(Zone_Temp_F > 77°F) = 14.155% (lab A4 from Week 6). Even though the all-hours r is 0.75, the zone temperature exceeds 77°F in 14% of hours — predominantly non-business hours when HVAC is in setback mode. The tail risk (14%) is meaningful even with a moderate correlation.

Problem 3 — Reflection (12 pts)

Write a reflection of **100–150 words** using **at least two specific numbers** from your lab or assignment.

Prompt: Your colleague says: "The Pearson correlation between outdoor temperature and cooling energy is 0.78 — that's almost perfect. We should just use outdoor temperature to predict cooling load."

Using at least two specific numbers from your lab or assignment, explain both why this colleague is **partly right** AND **partly wrong**. Address:

1. What your r^2 from A5 tells you about how much variance is explained — and what the remaining unexplained variance represents.
2. At least one case where the outdoor-cooling correlation changes substantially (cite your B1–B4 seasonal results or your B7/B8 business-hours results). For example, compare your business-hours and non-business-hours r values from B7/B8 — they show that "0.78" is an average of very different subgroup relationships — not a single fixed number that applies everywhere.

Grading Summary

Problem	Description	Points
P1a	r(Outdoor, Cooling) for July only	10
P1b	r(Zone, Cooling) for business hours only	10
P1c	r^2(Zone, Cooling) all hours + written interpretation	10
P2a	r(Zone, Outdoor) for Summer and Winter (2 computations)	6
P2b	Written: interpret summer vs. winter r(Zone, Outdoor) gap	6
P2c	Written: why $r = 0.75$ is compatible with strong HVAC control	6
P3	Reflection paragraph, 100–150 words, 2 specific numbers	12
Total		**60**